

AI Powered Household Energy Consumption Prediction and Monitoring System

Capstone Project Step 1 Problem Understanding and Framing

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Purpose. This document defines the household electricity problem addressed by the capstone, translates it into a measurable data science task, and records the technical and business criteria used to evaluate the completed forecasting prototype. It distinguishes verified evidence from assumptions and clearly limits the system to consumption forecasting rather than official utility billing.

1 Background and Philippine Context

Electricity is an essential and widely used energy source in the Philippines. According to the Philippine Statistics Authority and Department of Energy 2023 Household Energy Consumption Survey, approximately 26.47 million Filipino households used electricity in 2023, equivalent to 94.8 percent of households. This was an increase from 18.28 million households in 2011 [1].

The financial effect of household energy use is especially relevant in Metro Manila. Preliminary results from the 2025 Family Income and Expenditure Survey show that housing, water, electricity, gas, and other fuels accounted for 27.3 percent of total family expenditure in the National Capital Region. This was the region's largest expenditure category, although the published percentage combines electricity with housing and other utilities [2].

Meralco is the principal electricity distributor in Metro Manila and surrounding provinces. Its franchise area covers 39 cities and 72 municipalities and serves more than eight million residential, commercial, and industrial customers [3].

1.1 Limited consumption visibility

A conventional Meralco meter continuously records electricity consumption in kilowatt-hours (kWh). However, the official meter reading used to prepare the bill is normally collected monthly. The household's monthly consumption is calculated as the difference between the current and previous meter readings [4].

Unless household members regularly inspect their meter or manually estimate appliance use, they may have limited visibility into accumulated consumption during the billing period. An increase may therefore be recognized only after much of the electricity has already been used or when the monthly bill becomes available. This is a consumer visibility problem, not an allegation that Meralco fails to measure or calculate bills correctly.

Meralco provides an Appliance Calculator that estimates hourly, daily, weekly, and monthly appliance consumption based on appliance type and reported usage duration. Meralco notes, however, that calculator results may differ from actual consumption, particularly for cooling appliances during warmer months [5]. This creates an opportunity for a data-driven system that learns from historical total consumption and warns households before projected use becomes unusually high.

1.2 Evidence summary

Evidence	Statistic	Relevance
Filipino households using electricity in 2023	26.47 million	Shows the number of households potentially affected
Share of households using electricity	94.8%	Establishes electricity as a widespread household concern

Evidence	Statistic	Relevance
Meralco customer base	More than 8 million	Shows the scale of relevance within its franchise area
NCR housing and utility expenditure	27.3%	Demonstrates the financial importance of housing and utilities
Conventional meter reading	Monthly	Supports the problem of delayed consumption visibility
Existing monitoring support	Estimated appliance use	Shows that support exists but is not an automatic forecast of total household use

1.3 Evidence limitation

No authoritative Philippine statistic was identified that establishes that most households do not monitor their electricity consumption. The project therefore does not present that claim as a verified fact. Instead, it uses the defensible observation that monthly meter reading can delay consumption visibility unless a household performs additional monitoring.

1.4 The 2026 bill shock issue

The need for earlier consumption visibility became more urgent in 2026, when sudden increases in electricity bills were widely described in public discussion as bill shock. A July 2026 report by the Philippine News Agency stated that Meralco's residential electricity rate rose from PHP 13.2730 per kWh in January to PHP 14.8261 per kWh in July, an increase of approximately 11.7 percent [6]. The July adjustment alone was PHP 0.3428 per kWh and was attributed mainly to higher fuel prices and an increase in the generation charge [7].

The issue also prompted proposed Senate inquiries into rising electricity rates, disputed meter readings, system losses, and whether existing regulations adequately protect consumers from excessive, unreasonable, or avoidable charges [6]. Separately, the Energy Regulatory Commission temporarily suspended collection of the PHP 0.0371 per kWh Green Energy Auction Allowance for May and June 2026 to help ease the financial burden on electricity consumers, later extending the suspension through July and August [8][9].

Higher bills can result from both greater electricity consumption and changes in the applicable rate. Meralco explains that household consumption commonly increases during summer as air conditioners, fans, and refrigerators operate longer or work harder, while changes in the generation charge can raise the bill even when a household is already practicing energy-saving measures [10].

This capstone addresses the consumption component of bill shock by forecasting daily kWh usage and evaluating early warnings when predicted consumption is unusually high. It does not predict regulatory charges or guarantee the final electricity bill. Any future cost estimate must use a clearly stated input rate and must be labeled as an estimate rather than an official Meralco billing calculation.

2 Business Problem

Many Philippine households have limited visibility into their accumulated electricity consumption before receiving their monthly bill. Although consumers can manually read their meters or estimate appliance usage, these approaches require regular effort and do not automatically predict future household consumption. As a result, households may identify unusually high usage only after much of the consumption has occurred, reducing the time available to adjust behavior and manage the expected bill.

The proposed system addresses this problem by using historical consumption data to:

- Predict upcoming household electricity consumption.
- Compare predicted usage with normal or historical consumption.

- Identify periods of unusually high predicted consumption.
- Warn users when predicted consumption may exceed a predefined threshold.
- Present consumption patterns and forecasts through a monitoring dashboard.

3 Data Science Problem

The primary data science task is time-series regression. Historical electricity observations and time-related features are used to predict a continuous numerical value.

Task-type justification. Regression is appropriate because electricity consumption is measured as a continuous numeric quantity in kWh. A classification task would only place usage into broad categories and would not provide the numerical forecast needed to estimate consumption or support household planning.

Target variable. Total household electricity consumption in kWh for the next day.

Forecast-horizon justification. A one-day horizon balances actionability and predictive reliability: it gives a household time to adjust discretionary appliance use while avoiding the greater uncertainty of a long-range forecast.

The implemented candidate predictors included:

- Previous daily consumption and lagged consumption values
- Rolling averages, minimums, maximums, variability, and recent consumption trends
- Day-of-week, month, season, weekend, and cyclical calendar indicators
- Voltage behavior and sub-metering features available in the source data
- Exponentially weighted consumption history and domain-derived change features

The forecast also supports a threshold-based monitoring rule that flags predicted consumption above the household's recent pattern. This alert remains a supporting application rule rather than a separate classification model.

3.1 Operational definition of high consumption

For model evaluation, a high-consumption day was defined as a day whose total consumption exceeded 120 percent of the rolling mean for the preceding 30 days. Only earlier observations were used to calculate the threshold, preventing future information from leaking into the prediction. The first 30 days were excluded from alert evaluation because they did not have a complete historical window. A deployed prototype may additionally allow a personal daily kWh limit, but the fixed rolling threshold was retained for consistent model comparison.

4 Problem Statement

This project develops and evaluates an AI-powered household energy consumption prediction and monitoring system that uses historical electricity data to forecast the next day's total energy consumption. It compares multiple regression models, selects the strongest model using chronological validation, and evaluates an early-warning rule for unusually high predicted consumption. The intended benefit is earlier visibility that can support informed household energy-management decisions, particularly in the Philippine and Meralco context. The prototype is not an official billing system.

5 Project Objectives

5.1 General objective

To develop and evaluate a machine-learning system that can accurately predict short-term household electricity consumption and support proactive energy monitoring.

5.2 Specific objectives

1. Analyze historical household electricity consumption patterns.
2. Prepare and transform time-series data into features suitable for predictive modeling.
3. Develop and compare at least two forecasting or regression models.
4. Evaluate the models using appropriate regression metrics and select the most reliable model.
5. Optimize the selected model through hyperparameter tuning.
6. Explain the variables that have the greatest influence on predicted consumption.
7. Generate warnings when predicted consumption exceeds a defined threshold.
8. Present historical consumption, forecasts, and warnings through an understandable interface.

6 Task Type

Component	Task type	Purpose
Consumption forecasting	Time-series regression	Predict next-day electricity consumption in kWh
Model comparison	Regression evaluation	Identify the most reliable forecast
High-consumption warning	Threshold monitoring	Warn when predicted usage exceeds normal or defined limits
Prediction explanation	Explainable AI	Show which variables influenced the forecast

The primary task type is regression because the target is a continuous numerical value measured in kWh.

7 Technical Success Metrics

Models were evaluated using chronological train, validation, and test periods rather than random shuffling, ensuring that later observations were not used to predict the past.

7.1 Primary metric

Mean Absolute Error (MAE) will be the primary evaluation metric because it reports the average absolute prediction error in the same unit as the target, making the result easy to interpret in kWh.

$$\text{MAE} = (1/n) \times \sum |\text{actual consumption} - \text{predicted consumption}|$$

7.2 Supporting metrics

Metric	Purpose
MAE	Measures the average absolute prediction error in kWh
RMSE	Gives greater weight to large prediction errors
MAPE	Expresses prediction error as a percentage
R ²	Measures how much variation in consumption is explained by the model

MAPE will be interpreted carefully because it becomes unreliable when actual consumption is zero or close to zero.

8 Technical Success Criteria and Final Outcome

The following criteria were established before final model selection. The completed evaluation shows which criteria were achieved and where further work remains necessary:

- Achieved: XGBoost produced lower MAE and RMSE than both persistence and seasonal-naive baselines.
- Achieved: test MAE improved by approximately 35 percent relative to the seasonal-naive baseline.
- Partially achieved: test MAPE was 18.73 percent rather than the aspirational 15 percent target; MAE remained the primary selection metric.
- Achieved: final performance was measured on 173 unseen, chronologically later daily observations.
- Achieved with caution: the train-validation comparison did not indicate classic overfitting, although temporal and geographic generalization risks remain.

The evaluated baselines used the most recent day's consumption and the equivalent day of the previous week. The seasonal-naive baseline was the stronger final benchmark. XGBoost achieved a test MAE of 3.7232 kWh compared with 5.7282 kWh for that baseline.

8.1 Rationale for the performance targets

The requirement for at least a 10 percent reduction in MAE ensures that the selected model provides a meaningful improvement over a simple persistence or seasonal-naive forecast rather than only a marginal numerical gain. The MAPE target of 15 percent or lower represents an initially acceptable household-level relative forecasting error while remaining demanding enough to support useful next-day decisions. MAE remains the deciding metric because it is directly interpretable in kWh and is less distorted by occasional large errors than RMSE.

9 Business Success Indicators

Because the capstone uses a public single-household dataset rather than a prospective trial with Meralco customers, business KPIs are defined as pilot measures. Forecast metrics can be reported now, while behavior change, avoided kWh, and peso savings require transparent simulations or future observation.

Business KPI	Measurement
Early-warning coverage	Percentage of high-consumption periods identified before they occur
Warning precision	Percentage of warnings followed by actual above-threshold consumption
Consumption visibility	Clear presentation of historical and predicted usage
Potential consumption reduction	Simulated reduction when users respond to an alert
Potential bill savings	Estimated savings from avoided kWh using a stated electricity rate
User action time	Hours or days of advance notice before predicted high usage

9.1 Proposed Business Targets and Current Status

- Supported by design: the one-day forecast horizon provides up to one day of advance visibility.
- Not achieved by the original warning rule: recall was 21.74 percent, detecting 5 of 23 high-consumption test days against the proposed 80 percent target.
- Nearly achieved but below target: warning precision was 71.43 percent against the proposed 75 percent target.
- Not yet validated: potential consumption reduction and peso savings require a prospective Philippine household pilot and must not be presented as realized benefits.

The forecasting objective was achieved, but the warning and business targets were not fully achieved. The result is therefore a promising forecasting prototype rather than a deployment-ready bill-shock alert system. Future application thresholds should be configurable and validated on local household data.

9.2 Alignment of technical metrics and business outcomes

Technical accuracy creates business value only when it improves the timing and reliability of household decisions. The following relationships will be evaluated explicitly.

Technical result	Business interpretation
Lower MAE	More accurate daily kWh estimates for household planning
Lower RMSE	Fewer large errors that could hide or exaggerate high-use days

Technical result	Business interpretation
High warning coverage	More actual high-consumption days identified in advance
High warning precision	Fewer unnecessary alerts and greater user trust
One-day forecast horizon	Enough time to adjust discretionary appliance use
Estimated avoided kWh	Transparent basis for simulated potential bill savings

10 Scope and Boundaries

10.1 Included

- Historical household electricity consumption analysis
- Data cleaning and time-series feature engineering
- Next-day energy consumption forecasting
- Comparison of at least two predictive models
- Hyperparameter tuning and model explainability
- High-consumption warnings and forecast visualization

10.2 Outside the initial scope

- Direct integration with Meralco smart meters
- Real-time control of household appliances
- Official calculation of a customer's Meralco bill
- Guaranteed reduction in electricity consumption
- Deployment across multiple utility providers
- Appliance-level recommendations when appliance-level data is unavailable

11 Step 1 Summary

Requirement	Definition
Business problem	Limited visibility into rising household electricity use before the monthly bill
Data science problem	Predict future electricity consumption from historical time-series data
Primary task	Time-series regression
Target variable	Next-day total household electricity consumption in kWh
Primary metric	Mean Absolute Error
Supporting metrics	RMSE, MAPE, and R^2
Baseline	Persistence and previous-week seasonal-naive forecasts; seasonal naive was the stronger final benchmark
Technical target	Target: at least 10% lower MAE than the stronger naive baseline; achieved: approximately 35% lower test MAE
High-consumption threshold	More than 120 percent of the preceding 30-day rolling mean
Business outcome	Earlier household consumption visibility; real behavior change and savings require prospective validation
Primary users	Household electricity consumers
Forecast horizon	One day ahead

References

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